

Machine Learning Exercise 3A

Problem 1 *For this exercise, you will use logistic regression to classify images from the Fashion MNIST dataset. This dataset contains grayscale images of 10 different types of clothing items, making it more challenging than the original MNIST digits dataset.*

The Fashion MNIST dataset (<https://github.com/zalando-research/fashion-mnist>) contains:

- 60,000 training images
- 10,000 test images
- Each image is 28×28 pixels (784 features when flattened)
- 10 classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot

You can load the dataset directly using `tensorflow.keras.datasets.fashion_mnist`.

- Load the Fashion MNIST dataset and explore its structure. Print the shapes of the training and test sets.*
- Display a single image and its corresponding label. You will need to visualize the 28×28 matrix as a grayscale image.*
- Display 100 random images from the dataset in a 10×10 grid. Use matplotlib to create a figure with the class name as title for each image.*
- Normalize the pixel values to the range $[0, 1]$ by dividing by 255.*
- Flatten the images from 28×28 to 784-dimensional vectors.*
- Create and train a one-vs-all logistic regression classifier using `LogisticRegression` from `scikit-learn` with `multi_class='ovr'`. Note: You may need to increase `max_iter` (e.g., to 1000) for convergence.*
- Compute and print the accuracy for training and test sets.*
- Display the confusion matrix for the test set. Which classes are most often confused with each other?*